

Dijkstra's algorithm



Dr. Md. Atiqur Rahman
Assistant Professor
Dept. of Computer Science & Engineering

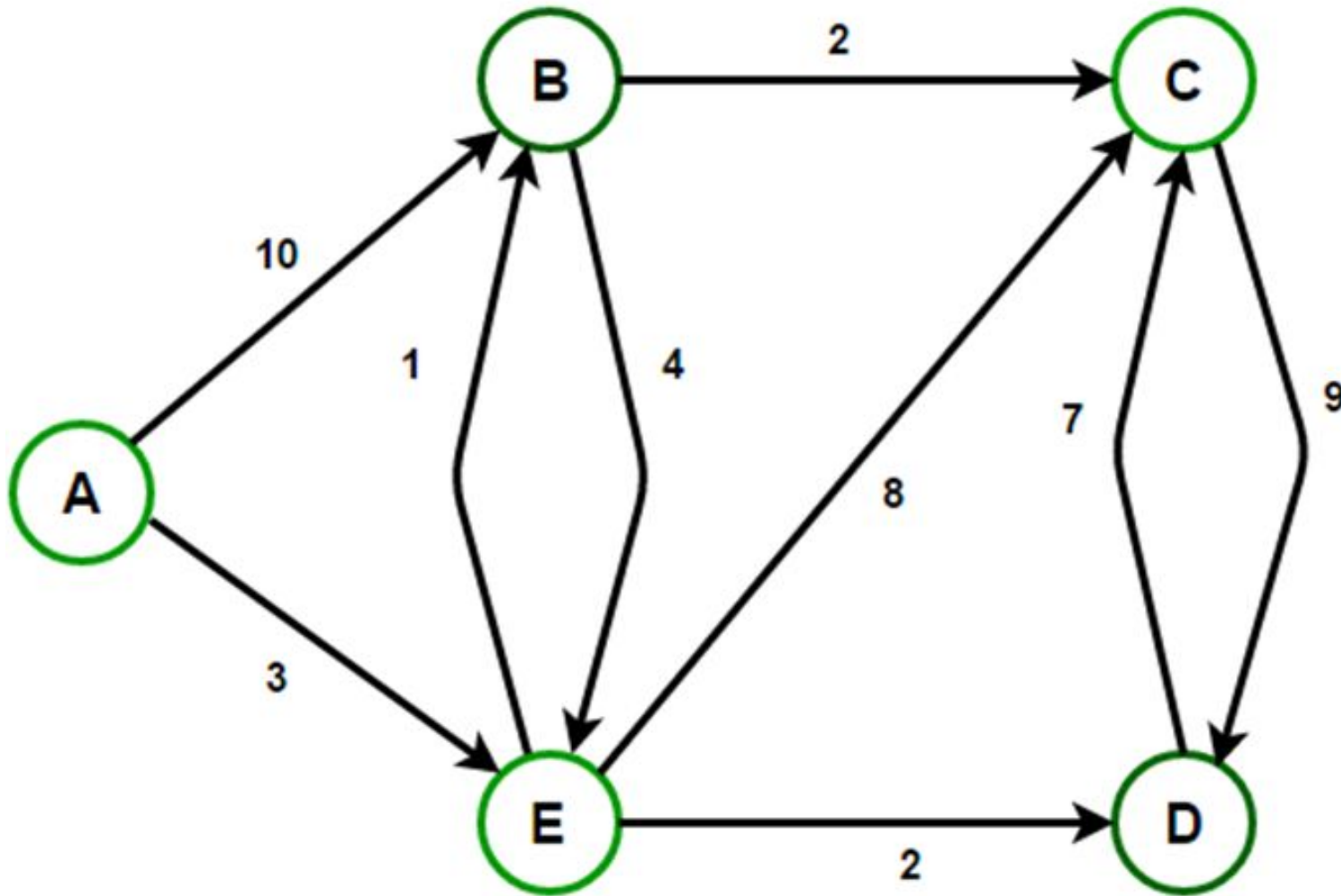
Single Source Shortest Path (Dijkstra's algorithm) - Greedy Method

Single Source Shortest Path (SSSP) is a graph algorithm that finds the shortest path from a single source node to all other nodes in a weighted graph. In other words, it determines the minimum cost to reach each vertex from a given starting vertex.

There are several algorithms to solve the Single Source Shortest Path problem, Such as

1. Dijkstra's algorithm: It is used to find the shortest path in a graph with non-negative edge weights.
2. Bellman-Ford algorithm: It is used to find the shortest path in a graph with negative or zero edge weights.

Single Source Shortest Path (Dijkstra's algorithm) - Greedy Method



What is relaxation?

Answer:

If $(d[u] + c(u,v)) < d[v]$
 $d[v] = d[u] + c(u,v)$

Selected	A	B	C	D	E
A	0	∞	∞	∞	∞

Thank You